PLASTIC WASTE MANAGEMENT IN THE UK

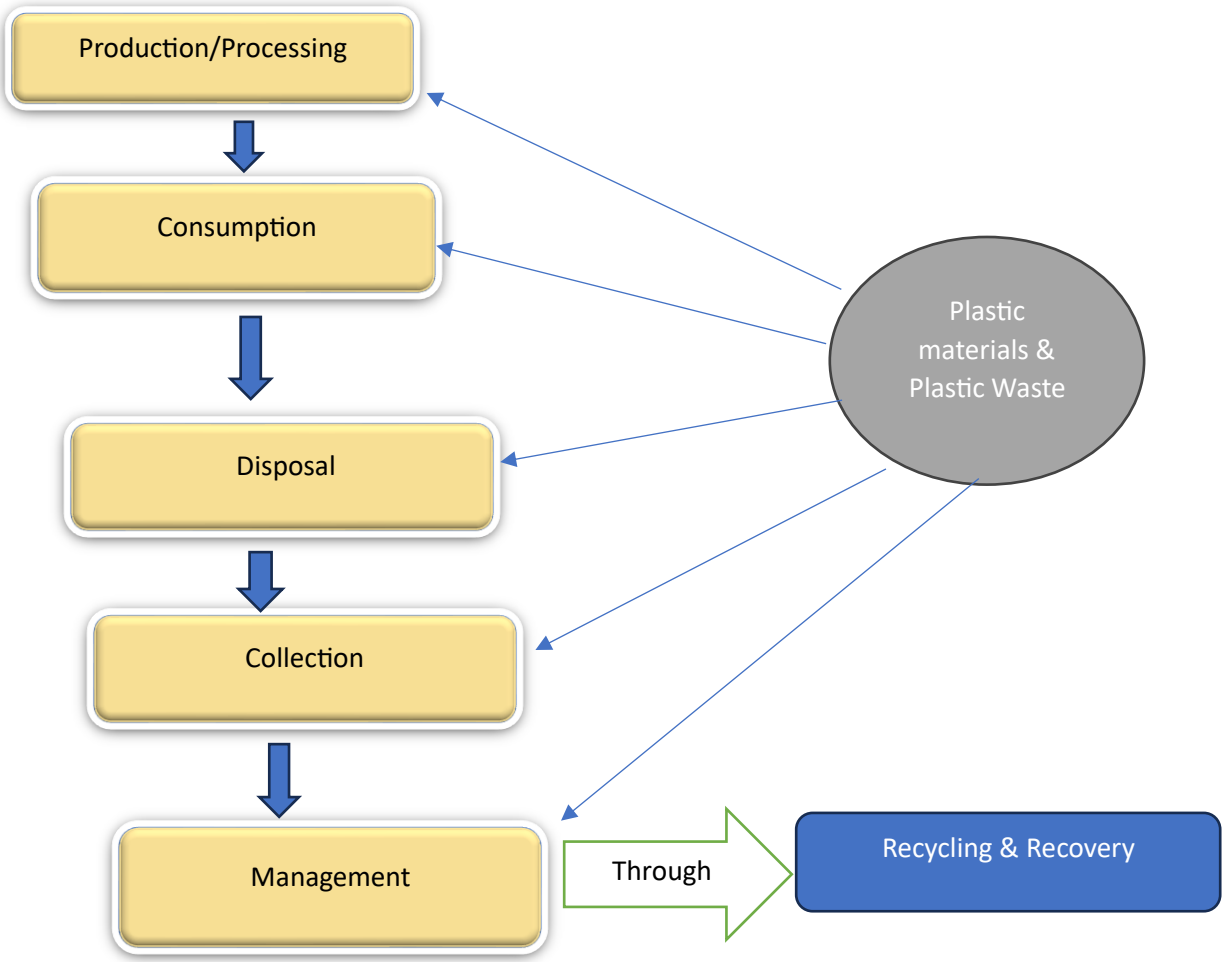
MARCH 1, 2024

PRINCE ASABI BOAKYE

Introduction

Plastic is a term used to represent a multi-array of lightweight organic polymers which are moldable, and used by plastic manufacturing industries. Thermoplastics, including polypropylene (PP), polystyrene (PS) etc, are usually obtained from fossil fuels [1]. Plastic waste is an important concern for the government of the United Kingdom (UK), having overwhelming statistics highlighting the intensity of the issue of plastic waste. As documented by the British Broadcasting Corporation (Blue Planet II), has increased the awareness of the need to tackle plastic waste holistically [1]. The government of the UK has drawn a good strategic measure to address the pressing issue of plastic waste. This is aimed to enhance the efficiency of resources. This is termed the Resources and Waste Strategy for England, which was introduced in December 2018 [2]. Households in the UK produce an estimated 100 billion pieces of plastic packaging every year [3]. Only 12% of plastic waste produced in the UK is domestically recycled, which leads to 21% of this waste being exported to countries with very poor waste management practices. It is important to increase the recycling process to reduce the emission of greenhouse gases (GHG) [4].

Flow Diagram



Production/Processing As of 2022, statistics estimated 0.5% of compostable plastic were produced out of all consumer plastic packaging products. Though the amount of plastic produced every year is decreasing, according to WRAP, there is a rise in the production of other packaging materials [5]. The UK government has recently changed the regulations governing the environment by introducing a plastic packaging tax, effective, 1st April 2024[6].

consumption In 2017, statistics showed an estimated 6.3 million tons of plastics consumed. An integral portion of these were imported as finished products [4]. The use of single-use plastic products (SUPP) was investigated in Portsmouth. Key factors that influence consumption are the availability of the product, inexpensiveness,

and convenience in its usage. 50% of all plastics in the UK are used for single-use disposable applications.[7] [8]

Collection & Disposal — In disposing of plastic waste, there should be a multi-bin collection system where there will be different companies contracted to collect the waste. Table 1 shows that there is a low collection rate in the UK. About 54% of rigid plastics are collected from households while the uncollected tonnage is dumped anyhow.[9]

Management — Drewniok et al discuss waste management in the UK and predict that there is a tendency for the UK to be able to manage 50% more plastic waste in 2050 than it does today. Plastic waste management in the UK is done through landfilling, incineration, and recycling. An approximated one-third, of each of plastic produced, goes into these technologies [4][10].

Recycling /Recovery — It is the basis of plastic waste management difficulties. Plastics are mostly incinerated in the UK. Plastic bottles are extracted and sent to recycling industries. Consumers are obliged to separate recyclables and non-recyclable plastics, domestically [9].

Table 1: Tonnages of plastics placed on the market (POM) in 2017 [9]
Source from: (WRAP,2018)

Stream	POM	Collected	Collection rate	Recycled	Exported for incineration
Consumer Total	1532k	525k	34%	226k	299k
Consumer PTTs/Bottles	*935k*	*509k*	*54%*		
Consumer Film	*395k*	*16k*	*4%*		
Consumer Other	*202k*	–	–		
Non-Consumer Total	830k	586k	–	199k	387k
Non-consumer Rigids	*453k*	*155k*	–	*55k*	*100k*
Non-consumer Film	*364k*	*431k*	–	*144k*	*287k*
Non-consumer Other	*13k*	–	–		
Grand Total	2361k	1111k	–	425k	686k
				40%	*60%*

Circularity Potential of Plastic Waste Management

Plastic waste is less recycled than other waste materials. This may be due to the multi-stage process involved. They can be chemically, mechanically, or thermally managed.[11]

UK's target of recycling 65% of waste established by the post-Brexit resources and Waste Strategy has failed. I only achieved between 45% and 47% of the set target. Lack of effective community engagements, social, economic, physical, human factors, and policy constraints have been the barriers to managing waste in the UK. Contributions from large plastic producers are currently scrutinized for their effectiveness. Parliament reports on the '*price of plastic'* aimed at 61% of recycling in 2022.[12]

The vision for a circular economy report may have some crucial challenges in its implementation and potentially prevent circularity achievements. There should be a long-term solution to prevent plastic waste from being left haphazardly in the environment.

Policy Change in the UK Towards Plastic waste Management

- A strategic ambition to work on plastic packaging on the market. This aims at making plastic waste reusable, compostable, and recyclable
- The government has introduced the plastic waste tax to increase revenue for its management. An updated target of 51% recycling in 2024 and 62% in 2030.
- There is a target to eliminate avoidable plastic waste by the year 2042. It is termed the *avoidable plastic waste*

The current Progress

The provisional report from 2021 data suggested that 44.2% of the set target was achieved. This rate has virtually been the same since 2015. However, other researchers think that the figures are less than what was achieved. There has been no data officially released on *avoidable plastic waste*. However, WRAP has researched and estimated a 46% reduction in plastic produced has been achieved since 2018 [5].

Proposed Solution for Sustainable Circularity of Plastic Waste

i. Advocation and calling for the implementation of the vision aligned in plastic management, thus, recycling, reusing, and recovering. This can be done through the collaboration of producers, retailers, and consumers.

ii. The extended producer responsibility (EPR) should be strengthened. This will let producers know that they are responsible for the entire life cycle of their product

iii. There should be a network and sharing of knowledge among researchers, chemists, and producers. This will promote an innovative way of finding solutions to the plastic waste menace

iv. There should be a strategic government ambition, like the Plastic Reduction and Action Plan, aimed at making plastics, recyclable, reusable, and compostable by 2025.

v. There should be a redesigning of products for a circular economy, where high priority is put on the end-of-life of the product. Again, the policymakers, the manufacturers, and the designers are the stakeholders involved

Conclusion

To be able to achieve the set target of the UK government, they should take important note of the key hindrances and address them accordingly. Some barriers were identified in this report. Among them, the physical factors, the lack of effective community engagement, and the political influence on the existing policies must be addressed. There should be deliberations in parliament on salient policies about plastic waste devoid of personal sentiments and political witch-hunting. The UK will be able to achieve its target and have a safe plastic waste-free environment if all targets are met

References

1. Grace McNicholas and Mathew Cotton, 2019, Stakeholder perceptions of Marine plastic waste management in the United Kingdom, vol 163, pp 77-87
2. Department for Environment Food & Rural Affairs, 2018, Resources and waste strategy: at a glance, 18th December 2018, https://www.gov.uk/government/publications/resources-and-waste-strategy-for-england/resources-and-waste-strategy-at-a-glance
3. Bruna Alves 2023, Plastic waste in the Uk-Statistics and Facts, 20th December 2023, https://www.statista.com/topics/4918/plastic-waste-in-the-united-kingdom-uk/#topicOverview
4. Drewniok Michal P et al., 2023, What to Do about Plastics? Lessons from a Study of United Kingdom Plastic flows, vol 57 pp 4513-4521
5. Environment, Food and Rural Affairs Committee, 2022, The price of Plastic: ending the toll of plastic waste, the third report of session 2022-23, https://committees.parliament.uk/publications/31509/documents/176742/default/
6. HM Revenue & Customs, 2023, Increase to Plastic Packaging Tax rates from 1 April, 2024, https://www.gov.uk/government/publications/changes-to-plastic-packaging-tax-rates-from-1-april-2024/increase-to-plastic-packaging-tax-rates-from-1-april-2024
7. Stephanie Lucy Northern et al, 2023, From shops to bins: a case study of consumer attitudes and behaviors towards plastics in a UK coastal city, vol 18, pp 1379-1395, https://doi.org/10.1007/s11625-022-01261-5
8. Jefferson Hopewell et al., 2009 Plastic recycling: challenges and opportunities, vol 364 pp 2115-2126 https://doi.org/10.1098%2Frstb.2008.0311
9. Martin Burgess et al, 2021, The future of UK plastics recycling: one Bin to Rule Them All, vol 164, https://doi.org/10.1016/j.resconrec.2020.105191
10. Jean-Paul Lange, 2021, Managing Plastic waste -Sorting, Recycling, Disposal and Product Redesign, vol 9 pp 15722-15738 https://doi.org/10.1021/acssuschemeng.1c05013
11. Ehsan Naderi Kalali et al., 2023, A critical review of the current progress of plastic was recycling technology in structural materials, vol 40 https://doi.org/10.1016/j.cogsc.2023.100763
12. Saeed Oluwadipe et al., 2021, A critical review of household recycling barriers in the United Kingdom, vol 40 pp 905-918, https://doi.org/10.1177/0734242X211060619